Customer Segmentation Analysis

Internship Project

Submitted By:

LIPSHA GIRI

Tools Used:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Google Colab

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler

sns.set(style="whitegrid")

print("Libraries Imported Successfully")
```

```
Libraries Imported Successfully
```

```
# Load dataset

df = pd.read_csv('ifood_df.csv')

# Display first 5 rows

df.head()
```

	Income	Kidhome	Teenhome	Recency	MntWines	MntFruits	MntMeatProducts	Mnt
0	58138.0	0	0	58	635	88	546	
1	46344.0	1	1	38	11	1	6	
2	71613.0	0	0	26	426	49	127	
3	26646.0	1	0	26	11	4	20	
4	58293.0	1	0	94	173	43	118	

5 rows × 39 columns

```
# Shape

print("Dataset Shape:", df.shape)

# Information

df.info()

# Missing values

print(df.isnull().sum())

# Columns

print(df.columns)
```

```
AcceptedCmp4            0
AcceptedCmp5            0
AcceptedCmp1            0
AcceptedCmp2            0
Complain                0
Z_CostContact           0
Z_Revenue               0
Response                0
Age                     0
Customer_Days           0
marital_Divorced        0
marital_Married         0
marital_Single          0
marital_Together        0
marital_Widow           0
education_2n Cycle      0
education_Basic         0
education_Graduation    0
education_Master        0
education_PhD           0
MntTotal                0
MntRegularProds         0
AcceptedCmpOverall      0
dtype: int64
Index(['Income', 'Kidhome', 'Teenhome', 'Recency', 'MntWines', 'MntFruits',
       'MntMeatProducts', 'MntFishProducts', 'MntSweetProducts',
       'MntGoldProds', 'NumDealsPurchases', 'NumWebPurchases',
       'NumCatalogPurchases', 'NumStorePurchases', 'NumWebVisitsMonth',
       'AcceptedCmp3', 'AcceptedCmp4', 'AcceptedCmp5', 'AcceptedCmp1',
       'AcceptedCmp2', 'Complain', 'Z_CostContact', 'Z_Revenue', 'Response',
       'Age', 'Customer_Days', 'marital_Divorced', 'marital_Married',
       'marital_Single', 'marital_Together', 'marital_Widow',
       'education_2n Cycle', 'education_Basic', 'education_Graduation',
       'education_Master', 'education_PhD', 'MntTotal', 'MntRegularProds',
       'AcceptedCmpOverall'],
      dtype='object')
```

```
# Remove duplicates

df.drop_duplicates(inplace=True)

# Fill missing values

df.fillna(method='ffill', inplace=True)

print("Data Cleaning Completed")
```

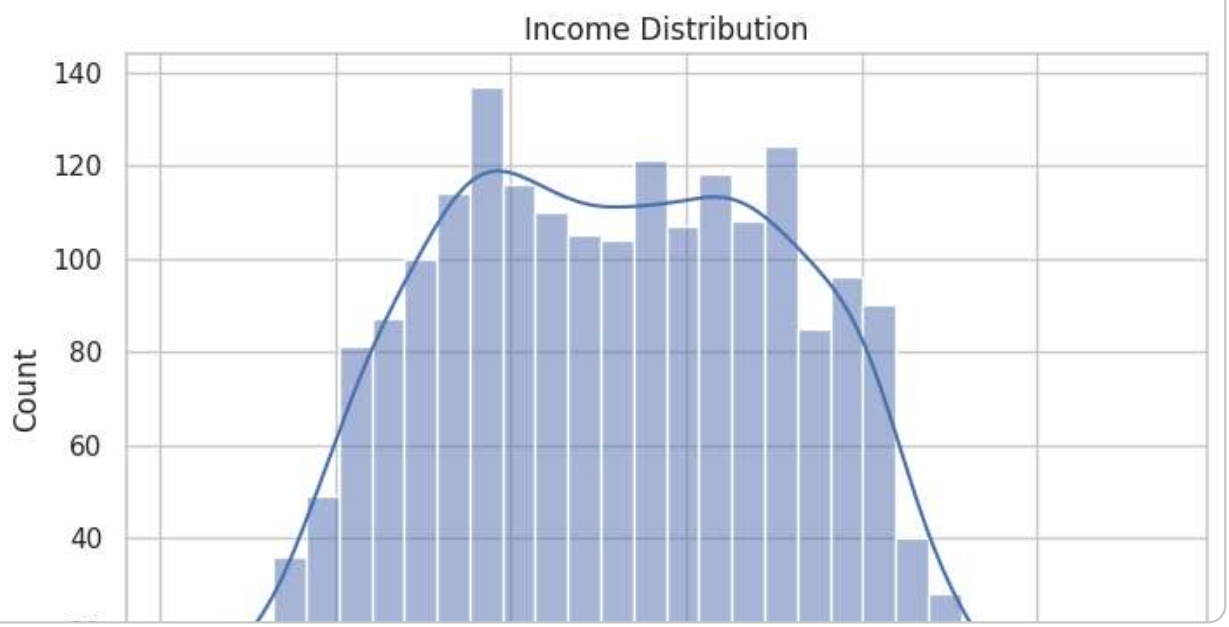
```
Data Cleaning Completed
/tmp/ipykernel_9710/1583781570.py:7: FutureWarning: DataFrame.fillna with 'metho
  df.fillna(method='ffill', inplace=True)
```

```
# Statistical summary
```

```
print(df.describe())
```

```
              Income      Kidhome     Teenhome      Recency     MntWines  \
count    2021.000000  2021.000000  2021.000000  2021.000000  2021.000000
mean    51687.258783     0.443345     0.509649    48.880752   306.492331
std     20713.046401     0.536196     0.546393    28.950917   337.603877
min      1730.000000     0.000000     0.000000     0.000000     0.000000
25%     35416.000000     0.000000     0.000000    24.000000    24.000000
50%     51412.000000     0.000000     0.000000    49.000000   178.000000
75%     68274.000000     1.000000     1.000000    74.000000   507.000000
max    113734.000000     2.000000     2.000000    99.000000  1493.000000

         MntFruits  MntMeatProducts  MntFishProducts  MntSweetProducts  \
count  2021.000000      2021.000000      2021.000000       2021.000000
mean     26.364671       166.059871        37.603662         27.268679
std      39.776518       219.869126        54.892196         41.575454
min       0.000000         0.000000         0.000000          0.000000
25%       2.000000        16.000000         3.000000          1.000000
50%       8.000000        68.000000        12.000000          8.000000
75%      33.000000       230.000000        50.000000         34.000000
max     199.000000      1725.000000       259.000000        262.000000

       MntGoldProds  ...  marital_Together  marital_Widow  education_2n Cycle  \
count   2021.000000  ...       2021.000000    2021.000000         2021.000000
mean      43.921821  ...          0.251856       0.034636            0.090549
std       51.678211  ...          0.434186       0.182902            0.287038
min        0.000000  ...          0.000000       0.000000            0.000000
25%        9.000000  ...          0.000000       0.000000            0.000000
50%       25.000000  ...          0.000000       0.000000            0.000000
75%       56.000000  ...          1.000000       0.000000            0.000000
max      321.000000  ...          1.000000       1.000000            1.000000

       education_Basic  education_Graduation  education_Master  education_PhD  \
count      2021.000000           2021.000000       2021.000000    2021.000000
mean          0.024245              0.502227          0.165760       0.217219
std           0.153848              0.500119          0.371957       0.412455
min           0.000000              0.000000          0.000000       0.000000
25%           0.000000              0.000000          0.000000       0.000000
50%           0.000000              1.000000          0.000000       0.000000
75%           0.000000              1.000000          0.000000       0.000000
max           1.000000              1.000000          1.000000       1.000000

          MntTotal  MntRegularProds  AcceptedCmpOverall
count  2021.000000      2021.000000         2021.000000
mean    563.789213       519.867392            0.302326
std     576.775749       554.797857            0.680812
min       4.000000      -283.000000            0.000000
25%      55.000000        42.000000            0.000000
50%     343.000000       288.000000            0.000000
75%     964.000000       883.000000            0.000000
max    2491.000000      2458.000000            4.000000

[8 rows x 39 columns]
```

```
plt.figure(figsize=(8,5))

sns.histplot(df['Income'],
             bins=30,
             kde=True)

plt.title('Income Distribution')

plt.show()
```



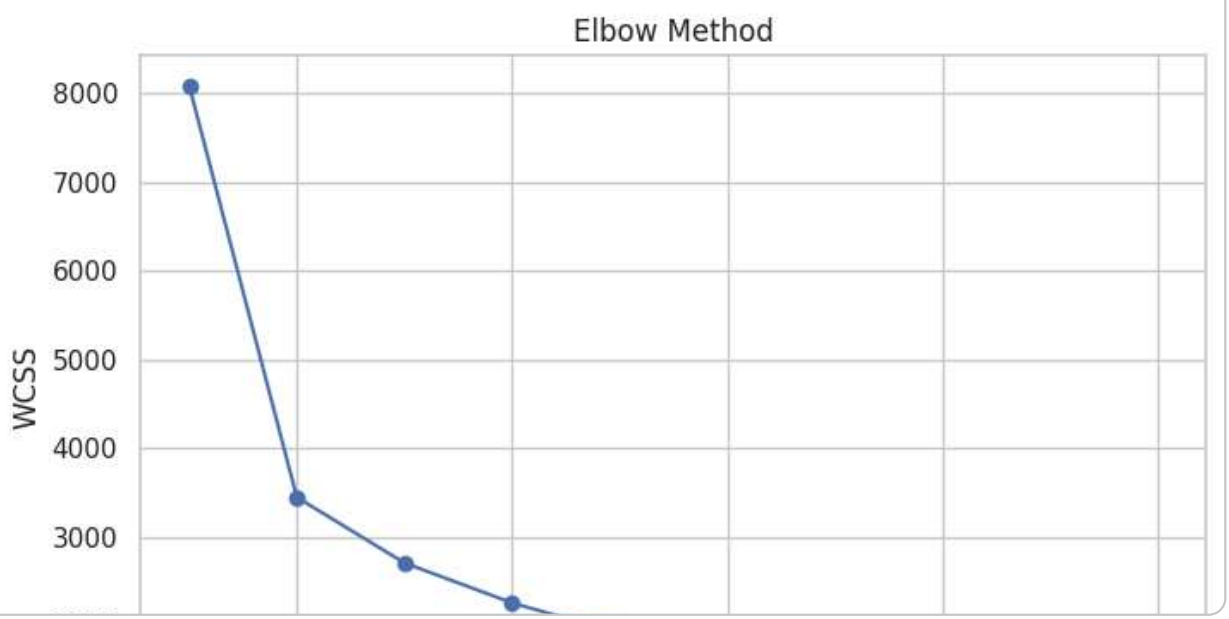

```
plt.figure(figsize=(8,5))

sns.histplot(df['MntTotal'],
             bins=30,
             kde=True)

plt.title('Total Spending Distribution')

plt.show()
```

```
plt.figure(figsize=(8,5))

sns.histplot(df['Age'],
             bins=20,
             kde=True)

plt.title('Customer Age Distribution')

plt.show()
```

```
# Feature selection

X = df[['Income',
        'MntTotal',
        'NumWebPurchases',
        'NumStorePurchases']]
```

```
# Scaling

scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)
```

```
# Elbow Method

wcss = []

for i in range(1,11):

    kmeans = KMeans(
        n_clusters=i,
        random_state=42
    )

    kmeans.fit(X_scaled)

    wcss.append(kmeans.inertia_)

# Plot

plt.figure(figsize=(8,5))

plt.plot(range(1,11),
         wcss,
         marker='o')

plt.title('Elbow Method')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')

plt.show()
```




```
# Apply KMeans

kmeans = KMeans(
    n_clusters=4,
    random_state=42
)

df['Cluster'] = kmeans.fit_predict(X_scaled)

# Display rows

df.head()
```

	Income	Kidhome	Teenhome	Recency	MntWines	MntFruits	MntMeatProducts	Mnt
0	58138.0	0	0	58	635	88	546	
1	46344.0	1	1	38	11	1	6	
2	71613.0	0	0	26	426	49	127	
3	26646.0	1	0	26	11	4	20	
4	58293.0	1	0	94	173	43	118	

5 rows × 40 columns

```
plt.figure(figsize=(10,6))

sns.scatterplot(
    x=df['Income'],
    y=df['MntTotal'],
```

```
    hue=df['Cluster'],
    palette='deep',
    s=100
)

plt.title('Customer Segmentation')

plt.xlabel('Income')

plt.ylabel('Total Spending')

plt.show()
```

```
# Cluster summary

cluster_summary = df.groupby('Cluster').mean(numeric_only=True)

print(cluster_summary)
```

```
                Income   Kidhome  Teenhome    Recency    MntWines  MntFruits  \
Cluster
0        31939.842303  0.829787  0.433041  48.693367   25.712140   4.211514
1        64389.305714  0.151429  0.762857  49.531429  571.900000  43.288571
2        76928.353349  0.060046  0.217090  48.789838  683.362587  61.586605
3        52605.425968  0.350797  0.735763  48.792711  234.205011  18.451025

         MntMeatProducts  MntFishProducts  MntSweetProducts  MntGoldProds  \
Cluster
0              19.295369         6.016270          4.361702     13.637046
1             216.925714        58.525714         44.045714     71.351429
2             465.745958        88.745958         64.230947     73.466513
3              97.034169        27.970387         19.127563     48.031891

         ...  marital_Together  marital_Widow  education_2n Cycle  \
Cluster  ...
0        ...          0.251564       0.023780            0.116395
```

```
1        ...          0.277143       0.057143            0.082857
2        ...          0.221709       0.039261            0.073903
3        ...          0.261959       0.031891            0.066059

         education_Basic  education_Graduation  education_Master  \
Cluster
0               0.060075              0.479349          0.160200
1               0.002857              0.502857          0.140000
2               0.000000              0.524249          0.180139
3               0.000000              0.521640          0.182232

         education_PhD     MntTotal  MntRegularProds  AcceptedCmpOverall
Cluster
0             0.183980    59.596996        45.959950            0.086358
1             0.271429   934.685714       863.334286            0.371429
2             0.221709  1363.672055      1290.205543            0.787529
3             0.230068   396.788155       348.756264            0.161731

[4 rows x 39 columns]
```

Business Insights

Cluster 0

Customers in this segment have high income and high spending behavior. These are premium customers who contribute significantly to company revenue.

Cluster 1

Customers with low income and low spending behavior represent budget-conscious customers.

Cluster 2

Customers with moderate income and moderate spending show stable purchasing patterns and potential for future growth.

Cluster 3

Customers with high income but lower spending may require personalized offers and targeted promotions to improve engagement.

Recommendations

1. Loyalty Programs

Provide exclusive rewards and membership benefits for premium customers.

2. Personalized Marketing

Target customers with customized product recommendations based on purchasing behavior.

3. Promotional Campaigns

Offer discounts and deals to low-spending customers to increase engagement.

4. Improve Customer Retention

Focus on customer satisfaction through personalized communication and offers.

5. Data-Driven Strategies

Use customer segmentation insights to improve marketing and sales strategies.

Conclusion